Chapter 9. Factor Analysis

Model, estimation (principal components, principal factors, MLE), rotation, factor scores

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1 The Orthogonal Factor Model

The covariance of p observed variables is explained by a few, k, unobserved common factors.

Definition: factor model

$$X_i = \mu_i + \sum_{\ell=1}^{k} \lambda_{i\ell} F_\ell + U_i \quad (i = 1, \dots, p), \qquad \mathbf{X} = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{F} + \mathbf{U},$$

where $\boldsymbol{\Lambda} = (\lambda_{i\ell})$ is the $p \times k$ **loading matrix**, $\mathbf{F} \in \mathbb{R}^k$ the common factors and $\mathbf{U} \in \mathbb{R}^p$ the specific factors, with

$$\mathrm{E}(\mathbf{F}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{F}) = \mathbf{I}_k, \quad \mathrm{E}(\mathbf{U}) = \mathbf{0}, \quad \mathrm{Cov}(\mathbf{U}) = \boldsymbol{\Psi} = \mathrm{diag}(\psi_1, \dots, \psi_p), \quad \mathrm{Cov}(\mathbf{F}, \mathbf{U}) = \mathbf{0}.$$

Theorem: covariance structure

$$\mathrm{E}(\mathbf{X}) = \boldsymbol{\mu}, \qquad \mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}, \qquad \mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \boldsymbol{\Lambda},$$

$$\mathrm{Var}(X_i) = \underbrace{\sum_{\ell=1}^{k} \lambda_{i\ell}^2}_{h_i^2\text{: communality}} + \underbrace{\psi_i}_{\text{specific variance}}, \qquad \mathrm{Cov}(X_i, X_j) = \sum_{\ell} \lambda_{i\ell}\lambda_{j\ell} \ (i \neq j).$$

Proof. $\mathrm{Cov}(\boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}) = \boldsymbol{\Lambda}\mathrm{Cov}(\mathbf{F})\boldsymbol{\Lambda}^\top + \mathrm{Cov}(\mathbf{U}) + \boldsymbol{\Lambda}\mathrm{Cov}(\mathbf{F}, \mathbf{U}) + \mathrm{Cov}(\mathbf{U}, \mathbf{F})\boldsymbol{\Lambda}^\top = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$, and $\mathrm{Cov}(\mathbf{X}, \mathbf{F}) = \mathrm{Cov}(\boldsymbol{\Lambda}\mathbf{F} + \mathbf{U}, \mathbf{F}) = \boldsymbol{\Lambda}$. ■

Because $\boldsymbol{\Psi}$ is diagonal, covariances between variables come *entirely* from the common factors. The model explains the $p(p+1)/2$ entries of $\boldsymbol{\Sigma}$ with $pk + p$ parameters.

1.1 Non-identifiability and rotation

For orthogonal $\mathbf{T}$ $(k \times k)$, $\boldsymbol{\Lambda}^* = \boldsymbol{\Lambda}\mathbf{T}$ and $\mathbf{F}^* = \mathbf{T}^\top\mathbf{F}$ give $\boldsymbol{\Lambda}^*\mathbf{F}^* = \boldsymbol{\Lambda}\mathbf{F}$, $\mathrm{Cov}\mathbf{F}^* = \mathbf{I}_k$ and $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top$: loadings are determined only up to rotation. If $\hat{\boldsymbol{\Lambda}}$ is an MLE, so is $\hat{\boldsymbol{\Lambda}}\mathbf{R}$ for any orthogonal $\mathbf{R}$. Removing the $k(k-1)/2$ rotational degrees of freedom leaves $pk + p - k(k-1)/2$ parameters and

$$\nu = \frac{p(p+1)}{2} - \left\{pk + p - \frac{k(k-1)}{2}\right\} = \frac{(p-k)^2 - p - k}{2},$$

which must be positive for the model to impose a real restriction.

1.2 Invariance to scale changes

If $Y_i = c_i X_i$, i.e. $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\mathbf{X}$, then $\mathbf{Y} = \mathrm{diag}(\mathbf{c})\boldsymbol{\mu} + \mathrm{diag}(\mathbf{c})\boldsymbol{\Lambda}\mathbf{F} + \mathrm{diag}(\mathbf{c})\mathbf{U}$ is again a factor model with $\boldsymbol{\Lambda}_Y = \mathrm{diag}(\mathbf{c})\boldsymbol{\Lambda}$ and $\boldsymbol{\Psi}_Y = \mathrm{diag}(\mathbf{c})^2\boldsymbol{\Psi}$. For standardised variables ($\mathrm{Var}X_i = 1$),

$$\lambda_{i\ell} = \mathrm{Cov}(X_i, F_\ell) = \mathrm{Corr}(X_i, F_\ell), \qquad \mathrm{Corr}(Y_i, F_\ell) = \lambda_{i\ell},$$

so loadings are read as variable–factor correlations. In practice factor analysis is often run on the correlation matrix $\mathbf{R}$.

2 Estimation

2.1 Principal component method

From $\widehat{\mathrm{Corr}}(\mathbf{X}) = \mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2} = \sum_{j=1}^{p} l_j \mathbf{u}_j \mathbf{u}_j^\top \approx \sum_{j=1}^{k} l_j \mathbf{u}_j \mathbf{u}_j^\top$ (largest k eigenvalues),

$$\hat{\mathbf{\Lambda}} = [\sqrt{l_1}\mathbf{u}_1, \ldots, \sqrt{l_k}\mathbf{u}_k], \qquad \hat{\mathbf{\Psi}} = \mathrm{diag}(\mathbf{R} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top).$$

The diagonal of $\mathbf{R}$ is reproduced exactly, and the sum of squared off-diagonal residuals of $\mathbf{R} - \hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top - \hat{\mathbf{\Psi}}$ is at most $l_{k+1}^2 + \cdots + l_p^2$. Factor ℓ explains a proportion l_ℓ/p of the total variance. Since eigenvalues do not change, increasing k leaves earlier loadings unchanged.

Remark: number of factors with the principal component method

Heuristics are used when PCA is used to estimate the factor loadings:

1. use a **scree plot** or a **cumulative scree plot** of $l_1 \geq l_2 \geq \cdots \geq l_p$;
2. or use **Kaiser's rule**: retain factors $1, \ldots, k$ satisfying

$$l_k > \bar{l} = \frac{1}{p}\sum_{i=1}^{p} l_i.$$

When correlation PCA is used, $\sum_i l_i = \mathrm{tr}\mathbf{R} = p$, so $\bar{l} = 1$ and the rule keeps factors with $l_k > 1$ (a factor must explain more than one standardised variable's worth of variance).

These are the same rules as for choosing the number of principal components (Ch. 7); the likelihood ratio test of §2.4 is available only under the normal MLE approach.

2.2 Principal factor method

Starting from initial communality estimates $\hat{h}_i^2$ (e.g. $1 - 1/r^{ii}$, r^{ii} the diagonal of $\mathbf{R}^{-1}$), form the reduced correlation matrix $\mathbf{R} - \hat{\mathbf{\Psi}}$ and iterate

1. $\mathbf{R} - \hat{\mathbf{\Psi}} \approx \hat{\mathbf{\Lambda}}_{(k)}\hat{\mathbf{\Lambda}}_{(k)}^\top$ (top-k eigendecomposition of the reduced matrix);
2. $\hat{\psi}_{(k)i} = 1 - \sum_{\ell=1}^{k} \hat{\lambda}_{(k)i\ell}^2$

until convergence.

2.3 Maximum likelihood

Assume $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})$. Profiling $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$, twice the negative log-likelihood is, up to constants and a factor n,

$$\ell(\hat{\boldsymbol{\mu}}, \mathbf{\Lambda}, \mathbf{\Psi}) = \log|\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi}| + \mathrm{tr}\{\mathbf{S}_0(\mathbf{\Lambda}\mathbf{\Lambda}^\top + \mathbf{\Psi})^{-1}\} + c.$$

For identifiability impose that $\mathbf{\Delta} = \mathbf{\Lambda}^\top\mathbf{\Psi}^{-1}\mathbf{\Lambda}$ is diagonal (if $\hat{\mathbf{\Lambda}}$ is an MLE, so is $\hat{\mathbf{\Lambda}}\mathbf{R}$).

Theorem: $\mathbf{\Lambda}$ given $\mathbf{\Psi}$

Given $\mathbf{\Psi} = \mathrm{diag}(\psi_1, \ldots, \psi_p)$, let $\mathbf{A} = \mathbf{\Psi}^{-1/2}\mathbf{S}_0\mathbf{\Psi}^{-1/2} = \sum_{i=1}^{p} d_i \mathbf{e}_i \mathbf{e}_i^\top$ ($d_1 \geq \cdots \geq d_p$) and $\delta_i = d_i - 1 > 0$ ($i \leq k$). Then

$$\hat{\mathbf{\Lambda}} = \mathbf{\Psi}^{1/2}[\mathbf{e}_1, \ldots, \mathbf{e}_k]\mathrm{diag}(\delta_1^{1/2}, \ldots, \delta_k^{1/2}).$$

Proof. Let $\boldsymbol{\Lambda}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Lambda}$ and $\boldsymbol{\Sigma}^* = \boldsymbol{\Psi}^{-1/2}\boldsymbol{\Sigma}\boldsymbol{\Psi}^{-1/2} = \boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} + \mathbf{I}_p$. Then $\ell = \log|\boldsymbol{\Psi}| + \log|\boldsymbol{\Sigma}^*| + \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1})$. If $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top}$ has eigenvalues $\gamma_1 \geq \cdots \geq \gamma_k \geq 0 = \cdots = 0$, $\boldsymbol{\Sigma}^*$ has eigenvalues $1+\gamma_j$ and 1. Von Neumann's lower bound (pair the small eigenvalues of $\boldsymbol{\Sigma}^{*-1}$ with the large ones of $\mathbf{A}$) gives

$$\log|\boldsymbol{\Sigma}^*| + \operatorname{tr}(\mathbf{A}\boldsymbol{\Sigma}^{*-1}) \geq \sum_{j=1}^{k}\Big\{\log(1+\gamma_j) + \frac{d_j}{1+\gamma_j}\Big\} + \sum_{j>k} d_j,$$

with equality when the eigenvectors of $\boldsymbol{\Sigma}^*$ coincide with the $\mathbf{e}_j$ in the same order. Each term $\log x + d/x = \log d + \{d/x - \log(d/x)\} \geq \log d + 1$ (Lemma 3.3) is minimised at $x = d$, so $1+\gamma_j = d_j$, i.e. $\gamma_j = \delta_j$. Hence $\boldsymbol{\Lambda}^*\boldsymbol{\Lambda}^{*\top} = \sum_{j\leq k}\delta_j\mathbf{e}_j\mathbf{e}_j^\top$ and $\boldsymbol{\Lambda}^* = [\mathbf{e}_1,\ldots,\mathbf{e}_k]\operatorname{diag}(\delta_j^{1/2})$ up to rotation; this choice makes $\boldsymbol{\Lambda}^{*\top}\boldsymbol{\Lambda}^* = \boldsymbol{\Delta}$ diagonal. ∎

Theorem: $\boldsymbol{\Psi}$ given $\boldsymbol{\Lambda}$

The MLE satisfies $\operatorname{diag}(\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}) = \operatorname{diag}(\mathbf{S}_0)$ (the model reproduces the sample variances). Accordingly, given $\boldsymbol{\Lambda}$, update (handwritten notes)

$$\hat{\boldsymbol{\Psi}} = \operatorname{diag}(\hat{\psi}_1,\ldots,\hat{\psi}_p), \qquad \hat{\psi}_i = (\mathbf{S}_0 - \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top)_{ii}.$$

Alternate the two steps. If some $\hat{\psi}_i \leq 0$ (a Heywood case), bound it below by a small positive number.

2.4 Likelihood ratio test for the number of factors

$H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$ (k factors) versus H_1: unrestricted $\boldsymbol{\Sigma}$. With $L_1 = L(\bar{\mathbf{x}}, \mathbf{S}_0)$ and $L_0 = L(\bar{\mathbf{x}}, \hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}})$,

$$W = -2\log\frac{L_0}{L_1} = n\log\Big(\frac{|\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top + \hat{\boldsymbol{\Psi}}|}{|\mathbf{S}_0|}\Big) \;\dot{\sim}\; \chi^2_\nu,$$

$$\nu = \frac{p(p+1)}{2} - \Big\{pk + p - \frac{k(k-1)}{2}\Big\} = \frac{(p-k)^2 - p - k}{2}.$$

(At the MLE $\operatorname{tr}\{\mathbf{S}_0\hat{\boldsymbol{\Sigma}}^{-1}\} = p$, so the trace terms cancel.) With Bartlett's correction, reject the k-factor model if $\big(n - 1 - \frac{2p+4k+5}{6}\big)\ln\frac{|\hat{\boldsymbol{\Lambda}}\hat{\boldsymbol{\Lambda}}^\top+\hat{\boldsymbol{\Psi}}|}{|\mathbf{S}_0|} > \chi^2_\nu(\alpha)$.

3 Factor Rotation

Non-identifiability is used to find interpretable loadings, ideally a simple structure in which each variable loads heavily on only one factor. Rotate $\boldsymbol{\Lambda}^* = \boldsymbol{\Lambda}\mathbf{J}$ with $\mathbf{J}$ orthogonal.

Definition: varimax and quartimax

Let $\tilde{\lambda}^*_{i\ell} = \lambda^*_{i\ell}/h_i$ (normalised by the communality).

$$\text{Varimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{\ell=1}^{k} \widehat{\operatorname{Var}}\big((\tilde{\lambda}^*_{1\ell})^2,\ldots,(\tilde{\lambda}^*_{p\ell})^2\big)$$

— spread the squared loadings within each *column* (factor), so each factor has some large and some near-zero loadings.

$$\text{Quartimax: } \mathbf{J}^\dagger = \arg\max_{\mathbf{J}} \sum_{i=1}^{p} \widehat{\operatorname{Var}}\big((\lambda^*_{i1})^2,\ldots,(\lambda^*_{ik})^2\big)$$

— spread the squared loadings within each *row* (variable), so each variable concentrates on one factor.

Rotation leaves the communalities h_i^2 and $\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top$ unchanged.

4 Factor Score Prediction

4.1 Weighted least squares (Bartlett) scores

(1) Choose $\hat{\mathbf{F}}$ to minimise the distance between $\mathbf{X}$ and $\boldsymbol{\mu} + \mathbf{\Lambda F}$: with the estimates fixed, regard $\mathbf{x}_i - \bar{\mathbf{x}} = \hat{\mathbf{\Lambda}}\mathbf{F}_i + \mathbf{U}_i$ as a heteroscedastic regression,

$$\hat{\mathbf{F}}_i^{\text{wls}} = \arg\min_{\mathbf{F}_i} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\mathbf{\Lambda}}\mathbf{F}_i)^\top \hat{\mathbf{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}} - \hat{\mathbf{\Lambda}}\mathbf{F}_i) = (\hat{\mathbf{\Lambda}}^\top \hat{\mathbf{\Psi}}^{-1} \hat{\mathbf{\Lambda}})^{-1} \hat{\mathbf{\Lambda}}^\top \hat{\mathbf{\Psi}}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$$

(set the gradient of the quadratic form to zero).

4.2 Regression scores

(2) Since $\begin{pmatrix} \mathbf{X} \\ \mathbf{F} \end{pmatrix} \sim N_{p+k}\Big(\begin{pmatrix} \boldsymbol{\mu} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{\Lambda\Lambda}^\top + \mathbf{\Psi} & \mathbf{\Lambda} \\ \mathbf{\Lambda}^\top & \mathbf{I}_k \end{pmatrix} \Big)$, the conditional expectation (Result 4.6) gives

$$\hat{\mathbf{F}}_i^{\text{reg}} = \mathrm{E}(\mathbf{F} \mid \mathbf{X} = \mathbf{x}_i) = \hat{\mathbf{\Lambda}}^\top (\hat{\mathbf{\Lambda}}\hat{\mathbf{\Lambda}}^\top + \hat{\mathbf{\Psi}})^{-1} (\mathbf{x}_i - \bar{\mathbf{x}}).$$

Theorem: relation between the two scores

With $\mathbf{\Delta} = \mathbf{\Lambda}^\top \mathbf{\Psi}^{-1} \mathbf{\Lambda}$, $\hat{\mathbf{F}}^{\text{reg}} = (\mathbf{I} + \mathbf{\Delta})^{-1} \mathbf{\Delta}\, \hat{\mathbf{F}}^{\text{wls}} = (\mathbf{I} + \mathbf{\Delta}^{-1})^{-1} \hat{\mathbf{F}}^{\text{wls}}$. If $\mathbf{\Delta}$ is diagonal each component is shrunk by $\delta_\ell/(1+\delta_\ell) < 1$, so $|\hat{F}_\ell^{\text{wls}}| \geq |\hat{F}_\ell^{\text{reg}}|$.

Proof. By Woodbury, $(\mathbf{\Lambda\Lambda}^\top + \mathbf{\Psi})^{-1} = \mathbf{\Psi}^{-1} - \mathbf{\Psi}^{-1}\mathbf{\Lambda}(\mathbf{I} + \mathbf{\Delta})^{-1}\mathbf{\Lambda}^\top \mathbf{\Psi}^{-1}$, so

$$\mathbf{\Lambda}^\top (\mathbf{\Lambda\Lambda}^\top + \mathbf{\Psi})^{-1} = \{\mathbf{I} - \mathbf{\Delta}(\mathbf{I} + \mathbf{\Delta})^{-1}\} \mathbf{\Lambda}^\top \mathbf{\Psi}^{-1} = (\mathbf{I} + \mathbf{\Delta})^{-1} \mathbf{\Lambda}^\top \mathbf{\Psi}^{-1}$$

(as $\mathbf{I} - \mathbf{\Delta}(\mathbf{I} + \mathbf{\Delta})^{-1} = (\mathbf{I} + \mathbf{\Delta} - \mathbf{\Delta})(\mathbf{I} + \mathbf{\Delta})^{-1}$), while $\mathbf{\Lambda}^\top \mathbf{\Psi}^{-1}(\mathbf{x} - \bar{\mathbf{x}}) = \mathbf{\Delta}\hat{\mathbf{F}}^{\text{wls}}$. ■

Regression scores shrink towards 0, reflecting the prior $\mathbf{F} \sim N(\mathbf{0}, \mathbf{I})$ (a Bayes estimate); WLS scores treat $\mathbf{F}$ as a fixed parameter and are conditionally unbiased.

5 PCA versus Factor Analysis

	PCA	Factor analysis
Goal	linear combinations explaining total variance $\mathrm{tr}\mathbf{\Sigma}$	latent factors explaining covariances (off-diagonal)
Model	none (descriptive transformation)	$\mathbf{\Sigma} = \mathbf{\Lambda\Lambda}^\top + \mathbf{\Psi}$
Direction	variables $\to$ components ($\mathbf{Z} = \mathbf{U}^\top \mathbf{X}$)	factors $\to$ variables ($\mathbf{X} = \boldsymbol{\mu} + \mathbf{\Lambda F} + \mathbf{U}$)
Uniqueness	unique if eigenvalues distinct	only up to rotation
Increasing k	earlier components unchanged	(MLE) earlier loadings change